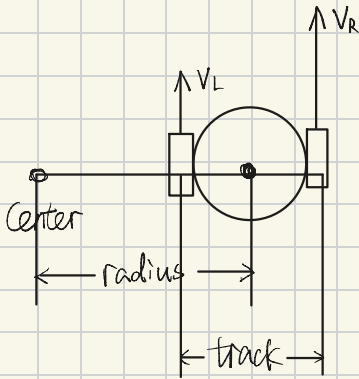


Forward Kinematics



$$\frac{V_L}{V_R} = \frac{\text{radius} - \frac{\text{track}}{2}}{\text{radius} + \frac{\text{track}}{2}}$$

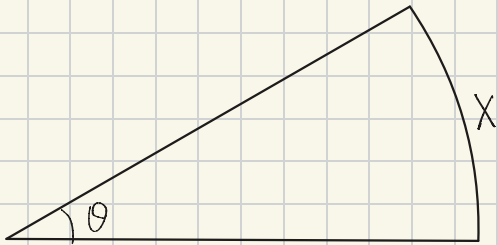
$$\Rightarrow \text{radius} = \frac{\text{track}(V_L + V_R)}{2(V_R - V_L)}$$

$$\Rightarrow \text{Twist. } x = \frac{V_L + V_R}{2}$$

$$\text{Twist. } y = 0$$

$$\text{Twist. } \theta = \frac{\text{Twist. } x}{\text{radius}} = \frac{V_R - V_L}{\text{track}}$$

Inverse Kinematics



$$\begin{cases} \text{Twist. } x = x \\ \text{Twist. } \theta = \theta \\ \text{Twist. } y = 0 \end{cases}$$

$$\text{radius} = \frac{x}{\theta}$$

$$\begin{cases} \frac{\text{radius} - \frac{1}{2}\text{track}}{\text{radius} + \frac{1}{2}\text{track}} = \frac{V_{\text{slower}}}{V_{\text{faster}}} \\ V_{\text{slower}} + V_{\text{faster}} = 2x \end{cases} \Rightarrow \begin{cases} V_{\text{slower}} = x - \frac{\text{track} \cdot x}{2\text{radius}} \\ V_{\text{faster}} = x + \frac{\text{track} \cdot x}{2\text{radius}} \end{cases}$$